REACT BCSL657B

Experiment 1

Use create-react-app to set up a new project. Edit the App.js file to include a stateful component with useState. Add an input field and a <h1> element that displays text based on the input. Dynamically update the content as the user types.

App.js

```
import React, { useState } from 'react';
import './App.css';

function App() {
    const [text, setText] = useState(''); // Declare a state variable 'text' and a function 'setText'
to update it
    const handleChange = (event) => {
      setText(event.target.value);
    }
   return (
      <div className="container" >
         <h1> Hello VCET</h1>
         <input
            type="text"
            onChange={handleChange} // Update state on input change
            placeholder="Type something..."

         />
         <h1>{text || "Type something..."}</h1>
      </div>
   );
}
export default App;
```

App.CSS

```
.container {
  text-align: center;
  margin-top: 50px;
}
```

Experiment 2

Develop a React application that demonstrates the use of props to pass data from a parent component to child components. The application should include the parent component named App that serves as the central container for the application. Create two separate child components**, Header**: Displays the application title or heading. **Footer**: Displays additional information, such as copyright details or a tagline. Pass data (e.g., title, tagline, or copyright information) from the App component to the Header and Footer components using props. Ensure that the content displayed in the Header and Footer components is dynamically updated based on the data received from the parent component.

Header.js (in Src)

```
import React from 'react';
function Header(props) {
 return (
 <header>
 <h1>{props.title}</h1>
 </header>
  );
 }
 export default Header;
```

Footer.js (in Src)

```
import React from 'react';
function Footer(props) {
   return (
      <footer>
         <p>{props.tagline}</p>
         <small>{props.copyright}</small>
      </footer>
   );
}
export default Footer;
```

App.js

```
import React from 'react';
```

```
import './App.css';

// Import the Header and Footer components
import Header from './Header';
import Footer from './Footer';

function App() {
   // Data to be passed to child components
   const title = "Welcome to React World";
   const tagline = "Code. Create. Innovate.";
   const copyright = "\u00A9 2026 Creative Coders Ltd.";

   return (
      <div className="App">
         {/* Pass data to Header and Footer components using props */}
         <Header title={title} />
         <Footer tagline={tagline} copyright={copyright} />
      </div>
   );
}
export default App;
```

App.css

```
.App {
   text-align: center;
}

header {
   background-color: slategray;
   color: white;
   padding: 20px;
}

footer {
   background-color: slategray;
   color: white;
   padding: 10px;
   position: fixed;
   width: 100%;
   bottom: 0;
 }
```